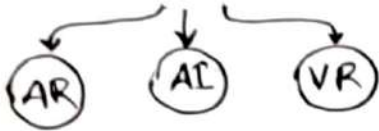


→ We need Human-level AI for Intelligent Assistant

Meta Glasses ☒ Not mainstream



→ Obstacles in AI development

→ Kulkashemi Company

↳ VLSI

→ The Ubiquitous AI Assistant is Becoming a Reality

↳ YOLO (You Only Look Once)

→ Machine Learning Sucks!

↓
Why?

- ① SL requires large labeled samples
- ② RL requires insane amounts of trials
- ③ SSL works great but generative prediction only works for text and other discrete modalities.

SL = Supervised Learning

RL = Reinforcement Learning

SSL = Self-Supervised Learning

☒ Chip crisis

→ Current evaluation of NVIDIA = 3.5T\$

→ Made using Rare earth materials

↳ Location? Russia — 1st
USA → Nevada State — 2nd
China — 3rd
(Tibet border)

Largest Oil Source?

Venezuela

↳ ☒ No technology to utilize that oil source

→ Why crisis?

↳ ☒ No proper production

→ NVIDIA sources from?

↳ Taiwan

Auto-Regressive LLMs

"Chain of Thoughts"

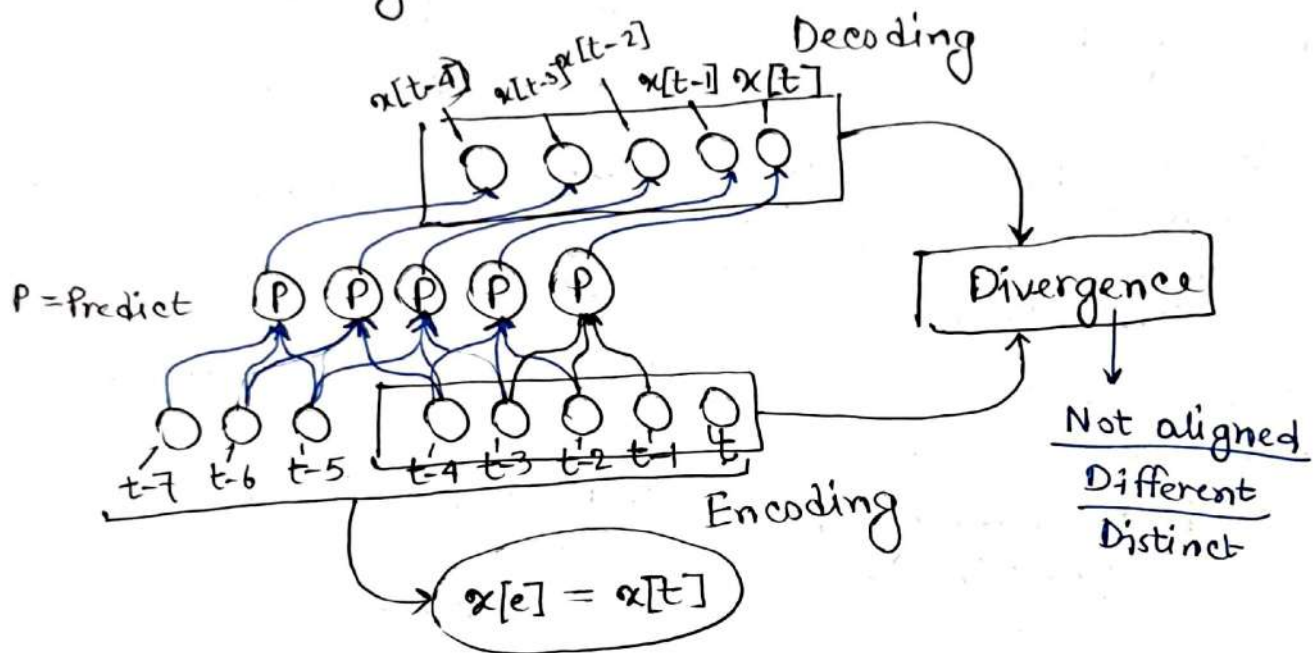
These are trained to predict the next word

How?

↳ Encoding

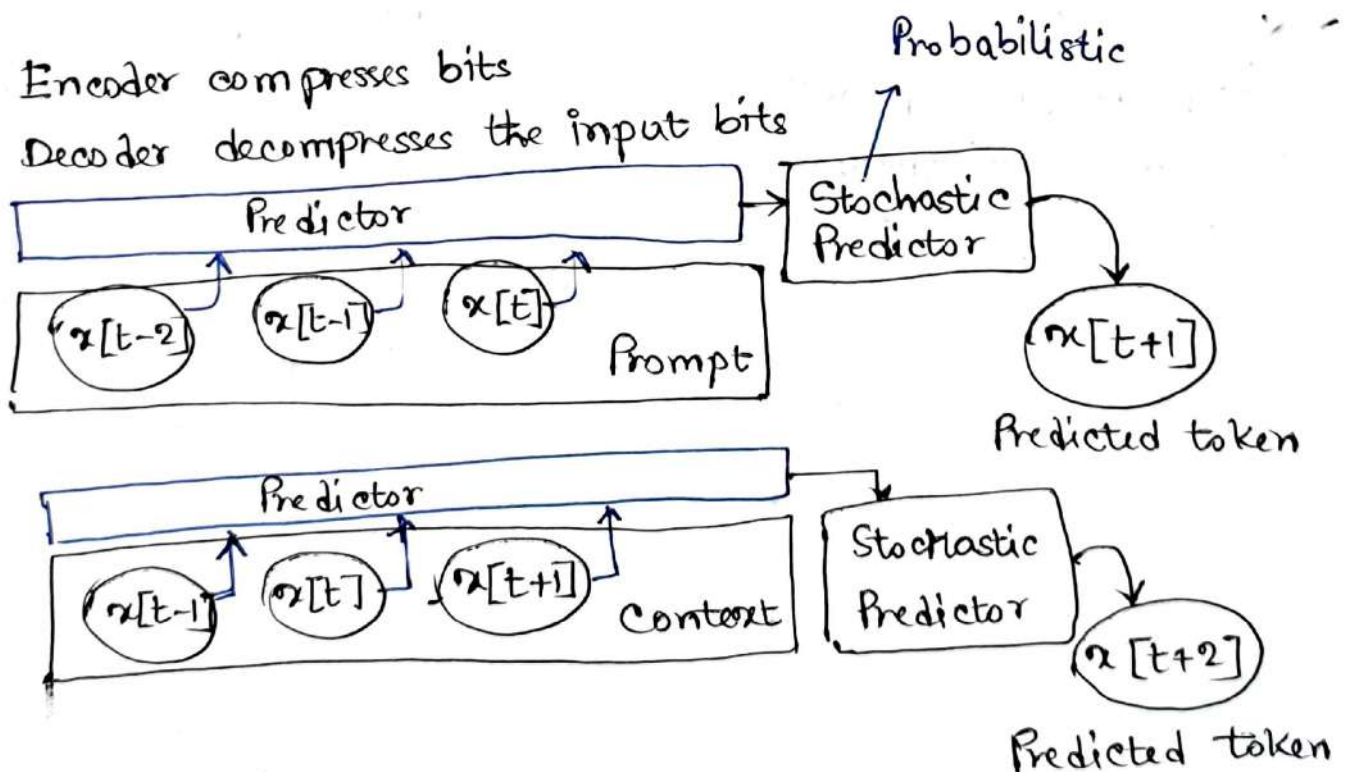
↳ Decoding

↳ Divergence



→ Encoder compresses bits

→ Decoder decompresses the input bits



Q. What is zero-shot?

Q. What is few-shot?

- ☒ No experience
- ☒ Given opportunity
- ☒ Some how good output
- ☐ Failed

By learning?

By luck?

→ Yeah, expected

☒ Some experience

→ * Either positive or negative outcome

"Moravec's Paradox"

↓
To test AI

→ How babies learn how the world works?

→ How do we get machines to learn like babies?

↓
"Reinforcement Learning"

→ Desiderata of AMI (Advanced Machine Intelligence)

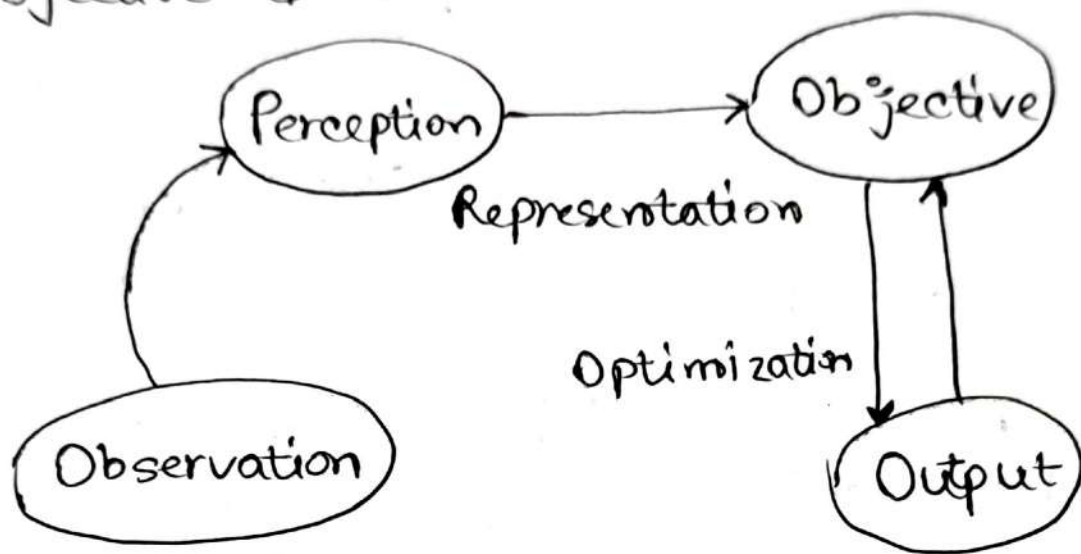
{Enigma} {Alan Turing} {Turing Test}

- ① Sensor-based Inputs
- ② Persistent Memory System
- ③ Plan of actions
- ④ Reasoning abilities
- ⑤ Safety & Control

{Current issue of AI: Uncertainty (অনিচ্ছয়তা)}

Shayan ①

→ Inference through optimization
Objective Driven AI



Ex. → Capturing dependencies with
Energy-based models

→ World Model for Planning/Reasoning